

Functional Equations

Note $\forall$ means "for all", $\mathbb{R}$ is the Real Numbers, $\mathbb{R}^+$ is the set of positive Reals.

Question difficulty is indicated by 🐣 (lower difficulty), 🐣🐣 (greater difficulty), and 🐣🐣🐣 (greatest difficulty). A topic label is also given on the question.

Rating: 🐣

Problem 1

Solve $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x+y)^2 = f(x)^2 + f(y)^2$, $\forall x, y \in \mathbb{R}$

Solution 1

Let $x = y = 0$. Then $f(0)^2 = 2f(0)^2$, so $f(0) = 0$.

Let $y = -x$. Then $0 = f(0)^2 = f(x)^2 + f(-x)^2$.

Since the sum of squares is non-negative, $f(x) = f(-x) = 0$.

Hence, $f(x) = 0$ for all $x \in \mathbb{R}$.

Problem 2

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x^2 + y) = f(x^{27} + 2y) + f(x^4)$.

Solution 2

Motivated by setting $x^2 + y = x^{27} + 2y$, let $y = x^2 - x^{27}$. Then $f(x^2 + y) = f(x^{27} + 2y)$, so $f(x^4) = 0$ for arbitrary x .

Hence $f(x) = 0$ for all non-negative reals. Now let $y = -x^2$, giving $0 = f(0) = f(x^{27} - 2x^2)$ for all real x . Since $x^{27} - 2x^2$ attains all real values, we conclude that $f \equiv 0$.

Problem 3

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f\left(\frac{x+f(x)}{2} + y\right) = f(x) + y$.

Solution 3

To make $\frac{x+f(x)}{2} + y = x$, let $y = \frac{x-f(x)}{2}$. Then

$$f(x) = f(x) + \frac{x - f(x)}{2},$$

so $f(x) = x$ for all x . This function can be verified to satisfy the equation, hence the unique solution is $f(x) = x$.

Problem 4

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, such that $x^2 f(x) + f(1-x) = 2x - x^4$.

Solution 4

Starting with

$$x^2 f(x) + f(1-x) = 2x - x^4 \quad (1)$$

we make the substitution $x \mapsto 1-x$, so that

$$(1-x)^2 f(1-x) + f(x) = 2(1-x) - (1-x)^4.$$

Thus,

$$f(1-x) = \frac{-f(x) + 2(1-x) - (1-x)^4}{(1-x)^2}.$$

Hence, from (1),

$$x^2 f(x) + \frac{-f(x) + 2(1-x) - (1-x)^4}{(1-x)^2} = 2x - x^4,$$

so

$$f(x) \left(\frac{(1-x)^2 x^2 - 1}{(1-x)^2} \right) = \frac{(1-x)^2 (2x - x^4) - 2(1-x) + (1-x)^4}{(1-x)^2},$$

which simplifies to $f(x) = 1-x^2$. It can be verified that this f satisfies the functional equation.

Problem 5

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(x) + y^2 + f(xy) = f(x+y)^2 - f(x)f(y)$$

for all real numbers x, y .

Solution 5

Exchanging the roles of x and y , we get

$$xf(x) + y^2 + f(xy) = f(x+y)^2 - f(x)f(y)$$

and

$$yf(y) + x^2 + f(xy) = f(x+y)^2 - f(x)f(y),$$

so that we obtain

$$xf(x) + y^2 = yf(y) + x^2,$$

meaning that

$$xf(x) - x^2 = yf(y) - y^2.$$

Since the LHS is independent of y and the RHS is independent of x , we must have

$$xf(x) - x^2 = C$$

for some constant C . Hence,

$$f(x) = x + Cx^{-1}.$$

Substituting this into the original equation, we have

$$x(x+Cx^{-1})+y^2+(xy+Cx^{-1}y^{-1}) = ((x+y) + C(x+y)^{-1})^2 - (x+Cx^{-1})(y+Cy^{-1}),$$

which yields $C = 0$. Hence, $f(x) = x$ is the only solution.

Rating: 🐼🐼

Problem 6

(USAMO 2002) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x^2 - y^2) = xf(x) - yf(y)$.

Solution 6

Solution here. here You can also finish this after obtaining that $f(x+y) = f(x) + f(y)$, $f(x^2) = xf(x)$ with

$$\begin{aligned} f((x+y)^2) &= (x+y)f(x+y) = (x+y)(f(x) + f(y)) \\ &= f(x^2) + f(y^2) + yf(x) + xf(y), \end{aligned}$$

and note $f((x+y)^2) = f(x^2 + y^2 + 2xy) = f(x^2) + f(y^2) + f(2xy)$, i.e., we have $f(2xy) = xf(y) + yf(x)$, where substituting $y = 1$ gives $f(2x) = 2f(x) = f(x) + f(1)x$, hence $f(x) = cx$ for constant c .

Problem 7

Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all positive integers x, y we have

$$2yf(f(x^2) + x) = f(x+1)f(2xy).$$

Solution 7

(BMO2 2022 P2) $x = 1$ to get $f(2y) = ky, k = \frac{2f(f(1)+1)}{f(2)}$ for some constant k , $y = 1 \implies f(2) = k, P(2z, 1) \implies f(2z+1) = kz + 1, P(2z+1, 1) \implies k = 2$, thus $f(x) = x$ for all $x \geq 2$, just use k to find $f(1) = 1$.

Problem 8

Let $\mathbb{R}^+$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x^2 + xf(y)) = f(f(x))(x+y)$$

for all positive real numbers x and y .

Solution 8

This is 2021 AMOC Senior Contest P5; see here.

Problem 9

(SMMC 2024 B2): Find all continuous functions $f : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ such that

$$f(x) = (x + 1)f(x^2),$$

for all $x \in \mathbb{R} \setminus \{-1, 1\}$.

Solution 9

For easy solving, we must transform it such that it relates a **clean** relationship involving $g(x), g(x^2)$ with no polynomial expression, where g is a transformation of f .

Substitute $g : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ with $g(x) = (x - 1)f(x)$. Then, by difference of squares, we have

$$g(x) = (x^2 - 1)f(x^2) = g(x^2).$$

for $x \in \mathbb{R} \setminus \{-1, 1\}$. From this equation, we also obtain $g(-x) = g(x)$, as $g(-x) = g(x^2)$.

Noting that $g(x) = g(x^2) = g(x^2^2) = \dots$, if x is between 0 and 1, then $g(x) = g(\lim_{n \rightarrow \infty} (x^2)^n) = g(0)$. For $x \in [0, 1)$, $\{x^{2^n}\}_{n=0}^{\infty}$ converges to 0, which means $g(x) = g(0)$ for $x \in [0, 1)$ by applying $g(x) = g(x^2)$ an arbitrarily large number of times.

For $x \in (-1, 0)$, $g(x) = g(0)$ because $g(-x) = g(x)$, which means the image of g on the $(0, 1)$ interval is equivalent to that on the $(-1, 0)$ interval.

Substituting $g(x^2) = g(x)$ with $x^2 = u$ and applying a dummy variable, we obtain $g(x) = g(x^{1/2}) = g(x^{1/2^2}) = \dots$. For $x > 1$, $\{x^{1/2^n}\}_{n=0}^{\infty}$ converges to 1, which means $g(x) = g(1)$ by applying $g(x^2) = g(x)$ an arbitrarily large number of times.

If $x > 1$, $\lim_{n \rightarrow \infty} x^{1/2^n} = x^0 = 1$, giving $g(x) = g(x^{1/2}) = \dots = g(\lim_{n \rightarrow \infty} x^{1/2^n}) = g(1)$. For $x < -1$, $g(x) = g(-1)$ because $g(-x) = g(x)$, which means the image of g on the $(-\infty, -1)$ interval is the same as that on the $(1, \infty)$ interval.

Since g is continuous except at $x = 1$, we expect a smooth horizontal line with open circles at $x = -1, 1$ (g is not defined at $x = -1, 1$) connecting g together.

This implies $g(x) = C$ for $x \in \mathbb{R} \setminus \{1\}$ where C is a real-valued constant.

Hence,

$$f(x) \in \left\{ p : \mathbb{R} \setminus \{-1, 1\} \rightarrow \mathbb{R}, C \in \mathbb{R} : p(x) = \frac{C}{x - 1} \right\}.$$

Problem 10

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(y)) + f(xy) = f(x + 1)f(y + 1) - 1.$$

Solution 10

We claim the solution is constant, $f \equiv c$, where $c^2 - 2c - 1 = 0$, or $f \equiv x$.

Note by symmetry we have $f(x + f(y)) = f(y + f(x))$. We suppose f is non-constant and attempt to prove injectivity.

We can get $f(a) = f(b) \implies f(a + 1) = f(b + 1)$ by $P(a, 1)$, and note with just putting a , we have

$$f(ax) = f(bx),$$

i.e., some real number r such that $f(rx) = f(x)$. Note we have, shifting $x \mapsto rx$ in the original,

$$f(x + 1) = f(rx + 1) = f(rx + r),$$

i.e., we now have a period $f(x) = f(x + p)$. Now shifting $y \mapsto y + p$, we have

$$f(xy) = f(xy + px),$$

i.e., f is constant, which is a contradiction. Now if f is injective, we have $f(x) = x + c$. Substitute back in to verify $c = 0$.

Rating: 🐾🐾🐾

Problem 11

Let $\mathbb{R}^+$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$xf(x^2)f(f(y)) + f(yf(x)) = f(xy)(f(f(x^2)) + f(f(y^2)))$$

for all positive real numbers x and y .

Solution 11

(2016 IMOSL A4)

Applying $P(1, 1)$ gives $f(1) = 1$. Further, from $P(x, 1)$ and $P(1, y)$, we get

$$f(f(y)) = f(y)f(f(y^2)) \implies f(x) = xf(x^2).$$

$$f(x)f(f(y)) + f(yf(x)) = f(xy) \left(\frac{f(f(x))}{f(x)} + \frac{f(f(y))}{f(y)} \right).$$

To prove injectivity, suppose a, b are distinct and $f(a) = f(b)$. Then we get $f(ay) = f(by)$, a contradiction to $f(x) = xf(x^2)$.

$$\begin{aligned} f(y)f(f(y^2)) &= f(f(y)) = f(y)f(f(y)^2) \\ \implies f(y^2) &= f(y)^2, \end{aligned}$$

which, together with $f(x) = xf(x^2)$, gives

$$f(x) = \frac{1}{x}.$$

Problem 12

(Putnam 2010 B5) Determine whether there exists a strictly increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the differential equation

$$f'(x) = f(f(x)) \quad \text{for all } x \in \mathbb{R}.$$

Solution 12

See the official Putnam archive [here](#).